

NHS GP Appointments: Data Cleaning & Missed Appointment Analysis

•**Tools:** Excel (Tables, PivotTables, Power Query)

•**Data:** Public NHS GP Appointments dataset (anonymised sample)



Business Questions I Wanted to Answer

- **Attendance rate:** Attended vs Cancelled vs Missed
- **Missed patterns:** Missed appointments by **hour/time band**
- **Operational quality:** How much data is incomplete (e.g., missing time)

Data cleaning was critical to ensure attendance rates and missed appointment patterns were calculated **accurately and consistently**.

Data Quality Issues Identified (Before Cleaning)

ISSUE	IMPACT
Patient_ID blanks (5)	Skews attendance rates
Appointment_Time blanks (12)	Breaks time analysis
Appointment_Type blanks (8)	Splits categories in pivots
Notes blanks (37)	Reduces reporting reliability
Inconsistent date formats	
Status inconsistency (No show / No-show)	
Doctor name inconsistency (Dr Jones / Dr. Jones)	

A	B	C	D	E	F	G	H	I
Patient_ID	Appointment_Date	Appointment_Time	Doctor_Name	Appointment_Type	Status	Notes		
	2025-01-01	08:00:00	Dr. Brown	Consultation	Cancelled	Late arrival		
	02-01-2025	13:15:00	Dr. Jones	Follow-up	Cancelled	Patient rescheduled		
X	03-01-2025	09:45:00	Dr. Patel	Emergency	Attended	Patient rescheduled		
	2025-01-04	08:00:00	Dr. Smith	Emergency	No-show	Late arrival		

Cleaning Steps (Excel + Power Query)

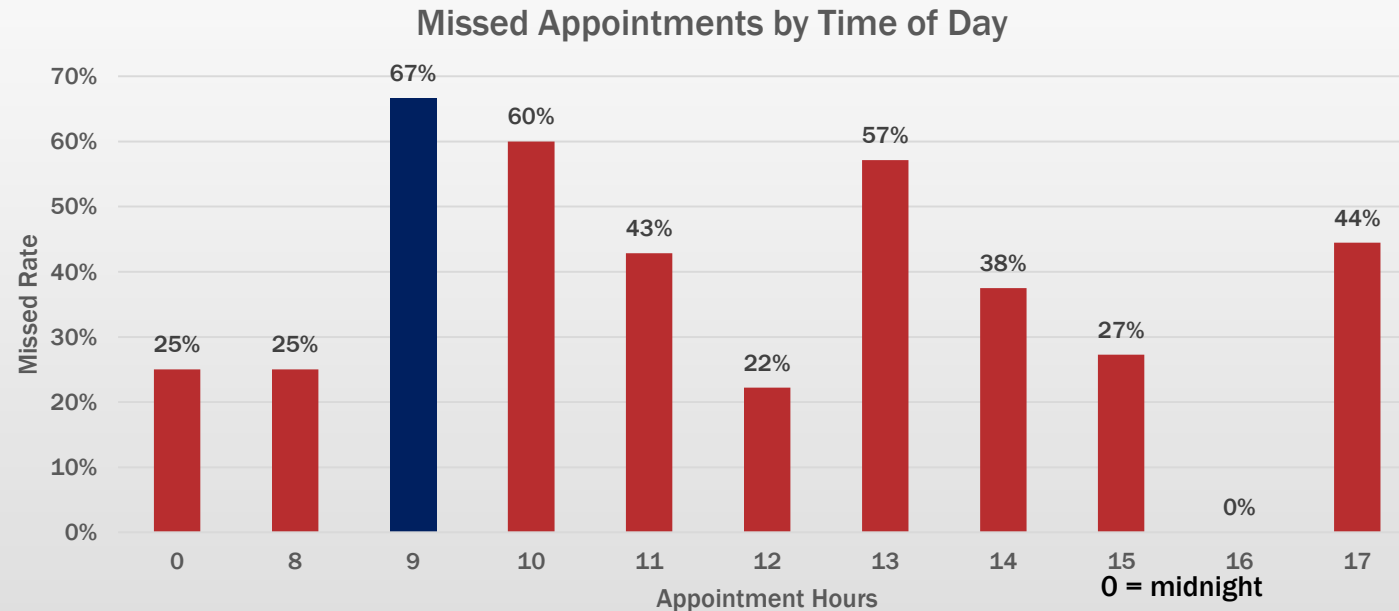
- Standardised **Appointment_Date** to yyyy-mm-dd (Power Query)
- Aligned **Appointment_Time** to 24-hour time format
- Cleaned text using **TRIM()** + **PROPER()**
- Standardised Doctor naming format (helper column)
- Standardised Status values (No show → Missed / No-show standard)
- Replaced blanks where needed: UNKNOWN, OMITTED NOTES
- Removed Patient_ID suffixes (e.g., P0003-X → P0003)
- Added **Missing_Time_Flag** (Yes/No)
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A	B	C	D	E	F	G	H	I	J
Patient_ID	Appointment_Time	Missing_Time_Flag	Doctor_Name	Appointment_Type	Appointment_Status	Missed_Appointm	Notes	Appointment_Hour	Appointment_Date
	8:00:00 AM	Yes	Dr Brown	Consultation	Cancelled		0 Late Arrival	8	2025-01-01
	1:15:00 PM	Yes	Dr Jones	Follow-Up	Cancelled		0 Patient Rescheduled	13	2025-01-02
	9:45:00 AM	Yes	Dr Patel	Emergency	Attended		1 Patient Rescheduled	9	2025-01-03
	8:00:00 AM	Yes	Dr Smith	Emergency	Missed		0 Late Arrival	8	2025-01-04
	12:45:00 PM	Yes	Dr Patel	Consultation	Cancelled		0 Late Arrival	12	2025-01-05
		No	Dr Jones	Emergency	Cancelled		0 Ommitted Notes	0	2025-01-06

After Cleaning: Dataset Ready for Analysis

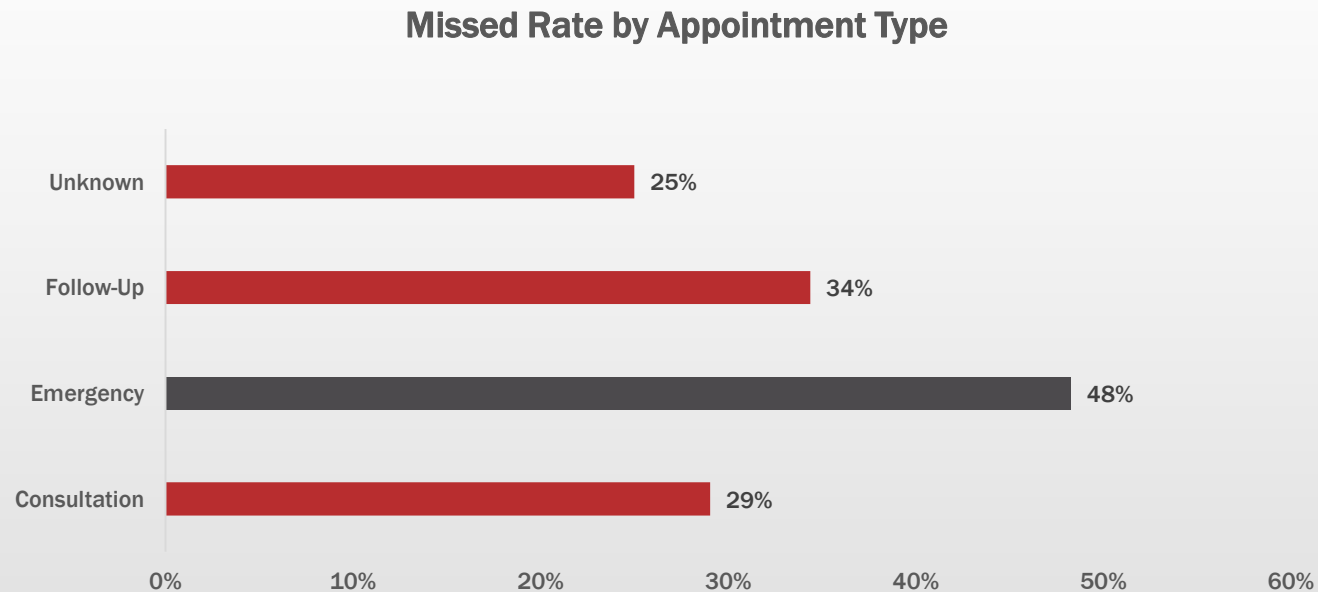
- Clean sample size: **100 rows | 10 columns**
- Created columns: **Appointment_Hour, Missing_Time_Flag, Missed_Appointment**
- Now possible to report:
 - Overall attendance rate
 - Missed rate by hour/time band
 - Missed rate by appointment type

Missed Appointments by Time of Day



✓ **Insight 1 — Time-based:** GP appointments scheduled at **9 AM** show the **highest missed rate (67%)**, suggesting early-morning slots may be more prone to non-attendance.

Missed Appointments by Appointment Type



✓ **Insight 2 — Type-based:** Emergency-labelled appointments have the highest missed rate (48%), which may indicate classification or booking issues rather than true emergency behaviour.

✓ **Insight 3 — Volume vs Risk:** Follow-up appointments have the highest volume, while emergency appointments carry the highest risk of being missed, highlighting the importance of separating volume from likelihood in operational planning.

Operational Implications

- Target reminders for **early-morning appointments**
- Review criteria for “**Emergency**” **appointment classification**
- Use missed-rate metrics alongside volume when planning capacity